

TUTORIAL NO 3

Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

```
#include <iostream>
using namespace std;

void swapElements(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
}

void inputArray(int arr[], int size) {
    cout << "Enter 5 elements:\n";
    for(int i = 0; i < size; i++) {
        cin >> arr[i];
    }
}

void displayArray(int arr[], int size) {
    cout << "Sorted array elements are:\n";
    for(int i = 0; i < size; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}

int main() {
    int arr[5];
    int n = 5;

    inputArray(arr, n);
```

```
for(int i = 0; i < n - 1; i++) {  
    for(int j = 0; j < n - i - 1; j++) {  
        if(arr[j] > arr[j+1]) {  
            swapElements(arr[j], arr[j+1]);  
        }  
    }  
}  
  
displayArray(arr, n);  
  
return 0;  
}
```

Output

Enter 5 elements:

2

4

1

7

9

Sorted array elements are:

1 2 4 7 9

=== Code Execution Successful ===